margins, poor differentiation, lympho-vascular invasion, and high tumor bud score. When none of these are present, it is determined to be curative resection (CR). Secondary outcome measures included surgery and adverse event (AE) rates. Peri-procedural predictors of NCR were also analyzed. Chi-square and Fisher exact tests were used where applicable. Statistical significance was set at a p-value of <0.05. SPSS v30.0.0.0 was used to perform all analyses. Results: 547 patients with mean age of 64 +/- 11.5 years were included. Proximal colon polyps (n=330, 60.3%) and right colon polyps (n=260, 47.5%) were well-represented. 25% of the polyps (n=135) had high-grade dysplasia and 66 patients (12.1%) had malignant polyps (Table 1). NCR for cancer was seen in 39 patients (7% of all polyps and 59% of all malignant polyps). CR for cancer was seen in 27 patients (5% of all polyps and 41% of malignant polyps. Prior polyp intervention, polyp location, size, circumference, morphology, presence of submucosal fibrosis, use of traction, hybrid ESD technique, age and procedure time were not significantly associated with NCR (Table 2). ESD-related AE were seen in 48 patients (8.8%). Around 8% of the patients (n=45) underwent surgery for AE, NCR or for other reasons. Conclusion: The majority (87.9%) of the colon polyps that underwent ESD were benign. Cancer CR with colon ESD was achieved in 27 patients (5% of all polyps). The rate of all-cause surgery and the rate of adverse events were both around 8%. There were no significant differences in NCR rates between proximal or right colon polyps to rest of the colon. These findings further question the role of colon ESD in its current form in the West. Strategies that will increase cancer CR rates, reduce surgery and AE rates will improve the risk-benefit profile of colon ESD.

Table 1. Prevalence of cancer, non-curative resections, ESD adverse events and surgery

Study variable	n (%)
High-grade dysplasia	137 (25%)
Malignant polyps	66(12.1%)
- Deep (>1 mm) submucosal invasion	- 35 (53%)
- Positive/Indeterminate margins	- 21 (31.8%)
- Lympho-vascular invasion	- 16 (24.2%)
- High tumor bud score	- 4 (6%)
Right colon malignant polyps (n=260)	27 (10.4%)
Proximal colon malignant polyps (prox. to splenic flexure) n=330	35 (10.6%)
Distal colon malignant polyps (distal to splenic flexure) n=217	31 (14.3%)
Non-curative resections for malignant polyps	
- n, % of all malignant polyps (n=66)	- 39 (59.1%)
- n, % of right colonmalignant polyps (n=27)	- 13 (48.1%)
- n, % of proximal colon malignant polyps (n=35)	- 18 (51.4%)
- n, % of distal colon malignant polyps (n=3l)	- 21 (67.7%)
ESD adverse events in all polyps	
{intraproceduraland delayed) n=547	48 (8.8%)
- Intraprocedural perforation	- 4 (0.7%)
- All delayed adverse events within 2 weeks	- 44 (8%)
- Delayed bleeding within 2 weeks	- 14 (2.6%)
- Delayed perforation within 2 weeks	- 15 (2.7%)
ESD adverse events in malignant polyps (n=66)	5 (7.6%)
- Intraprocedural perforation	- 0 (0%)
- All delayed adverse events within 2 weeks	- 5 (7.6%)
- Delayed bleeding within 2 weeks	- 0 (0%)
- Delayed perforation within 2 weeks	- 3 (4.5%)
Surgery	
- For all polyps (n-547)	- 45 (8.2%)
- For malignant polyps (n=66)	- 4 (6%)
- For complications (n=48)	- 8 (16.7%)
- For right colon polyps (n=260)	- 24 (9.2%)
- For proximal colon polyps (n=330)	- 31 (11.9%)

Table 2. Predictors for non-curative resection among malignant polyps (n=66)

Variable	Non-curative resection (n=39)	Curative resection (n=27)	p-value
Previous EMR or surgery	4 (10.3%)	4 (14.8%)	0.71
Proximal colon polyps	18 (46.2%)	15 (55.6%)	0.45
Right colon polyps	13 (33.3%)	13 (48.1%)	0.23
Paris 0-ls or lsp polyps	27 (69.2%)	14 (51.9%)	0.15
Traction used	10 (25.6%)	5 (18.5%)	0.50
Submucosal fibrosis	25 (64.1%)	13 (48.1%)	0.20
Hybrid ESD en-bloc resection	3 (9.4%)	1 (4.2%)	0.63
NICE3	8 (30.8%)	7 (31.8%)	0.94
JNET3	8 (20.5%)	4 (14.8%)	0.53
Age (mean, s.d.)	68 (10.2)	65.7 (10.2)	0.37
Circumferential (mean,)	35 (14.5)	38.9 (20.8)	0.44
Polyp size (mean, s.d.J	43.5 (21.7)	46.3 (26.5)	0.64
Procedure time (mean, s.d.J	131.5 (75)	144.4 (103)	0.50

ID: 4258940

ACCURACY ASSESSMENT OF VIRTUAL SCALE ENDOSCOPE FOR EARLY DETECTION OF COLORECTAL POLYPS: A META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

Umar Akram, Hamza Ashraf, Eeshal Fatima, Hareesha R. Bharadwaj, Faryal Altaf, George S. Zacharia, Franklin Sosa, Zaheer Qureshi

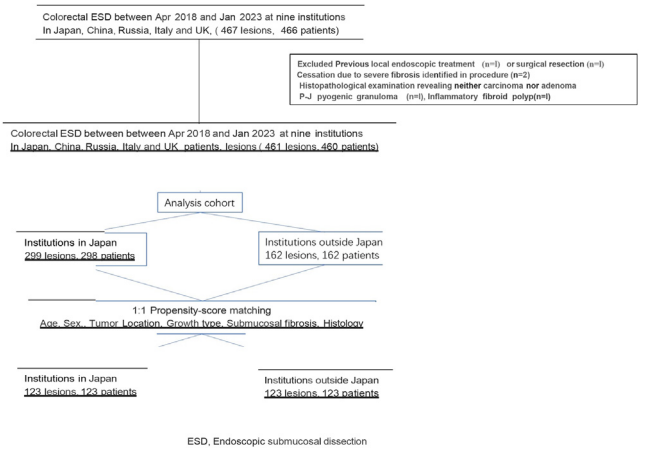
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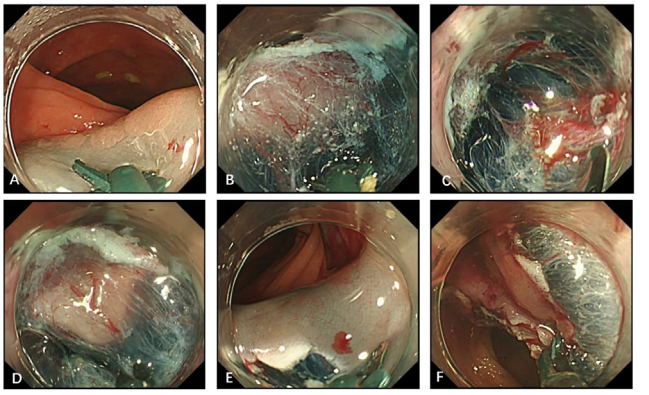
Abstract Body: BACKGROUND: Colonoscopy reduces colorectal cancer mortality by detecting and removing polyps, with polyp size critical for determining treatment and surveillance intervals. However, most endoscopists rely on subjective visual assessment (VA), which is prone to variability and inaccuracy. While uncalibrated tools like snares and forceps are often used, their impact on measurement accuracy is uncertain. The laser-based Virtual Scale Endoscope (VSE) integrates a precise size measurement scale and has shown superior accuracy in clinical studies. Aim: We conducted a systematic review and meta-analysis to evaluate the accuracy of VSE compared to VA and snares in measuring early-stage colorectal lesions. Methods: A thorough literature search was conducted on MEDLINE, Embase, and clinicaltrials.gov from inception till November 2024. Only randomized controlled trials (RCTs) comparing VSE with VA or snares and reporting relative accuracy were considered eligible. Cochrane risk-of-bias tool (RoB2) was used for bias assessment. Mean differences (MDs), risk ratios (RRs), and odds ratios (ORs) with 95% confidence intervals (CIs) were reported. Statistical analysis was conducted on R version 4.4.0. Results: A total of 9 RCTs involving 7828 polyps were included in this systematic review and meta-analysis. All the studies exhibited low to moderate risk of bias. Relative accuracy was significantly higher in the VSE group as compared to VA (RR, 1.22; 95% CI, 1.17 to 1.27; p<0.01; I^2=14%) and snares groups (RR, 1.25; 95% CI, 1.20 to 1.31; p<0.01; I^2=0%). VSE underestimated polyp size significantly lower than VA (RR, 0.49; 95% CI, 0.25 to 0.95; p=0.04; I^2=94%). However, results were comparable when compared with the snare group (RR, 0.93; 95% CI, 0.76 to 1.14; p=0.49; I^2=0%). VSE overestimated polyp size lower than VA (RR, 0.74; 95% CI, 0.47 to 1.17; p=0.20; I^2=80%) and snare groups (RR, 0.66; 95% CI, 0.54 to 0.80; p<0.01; I^2=0%). Overall analysis revealed that VSE misclassified poly size less when compared with VA and snare groups. Misclassification results are detailed in Table 1. VSE is associated with a higher average time for size estimation of polyps as compared to VA (MD, 24.34; 95% CI, 22.07 to 26.60; p<0.01; I^2=0%). Conclusion: This meta-analysis highlights the superior accuracy of the VSE compared to traditional VA and uncalibrated tools like snares in measuring colorectal lesion sizes. VSE significantly reduces the likelihood of both underestimation and overestimation, ensuring more precise polyp classification. This precision is crucial for guiding appropriate treatment and surveillance intervals, ultimately improving patient outcomes. Adopting VSE in routine clinical settings could address the limitations of subjective assessments, making it a valuable advancement in colorectal cancer prevention and management.

Table 1: Polyp size misclassification, comparison of virtual scale endoscope and virtual assessment

Misclassification: VSE versus VA				
Polyp size	**OR**	**95% CI**	**P value**	**I²**
≤3mm misclassified as >3mm	0.94	0.36 to 2.46	0.90	0%
>3mm misclassified as ≤3mm	0.23	0.04 to 1.28	0.09	83%
≤5mm misclassified as >5mm	0.54	0.30 to 0.98	0.04*	56%
>5mm misclassified as ≤5mm	0.24	0.13 to 0.45	<0.01*	32%
≤10mm misclassified as >10mm	1.08	0.57 to 2.02	0.82	0%
>10mm misclassified as ≤10mm	0.39	0.22 to 0.69	<0.01*	12%
≥20mm misclassified as <20mm	0.31	0.18 to 0.52	<0.01*	0%
< 20mm misclassified as ≥20mm	6.57	1.66 to 25.95	<0.01*	0%
Misclassification: VSE versus Snares				
≤5mm misclassified as >5mm	0.64	0.27 to 1.56	0.33	58%
>5mm misclassified as ≤5mm	0.15	0.03 to 0.69	0.01*	76%
≤10mm misclassified as >10mm	0.65	0.15 to 2.76	0.56	75%
>10mm misclassified as ≤10mm	2.22	0.41 to 12.18	0.36	0%

Abbreviations: VSE, virtual scale endoscope; VA, visual assessment; OR, odds ratio; CI, confidence intervals; I², heterogeneity; *, statistically significant





ID: 4249618

THE SAFETY AND EFFICIENCY OF SCISSOR-TYPE KNIVES IN COLORECTAL ESD:AN INTERNATIONAL MULTICENTER PROSPECTIVE OBSERVATIONAL STUDY

Kuilang Liu, Jing Wu, Yuzuru Tamaru, Yadan Wang, Hui Su, Chunmei Guo, Canghai Wang, Hong Liu, makoto kobayashi, Kiyoaki Homma, Takuya Yamada, Levchenko Evegeniv, Noor Mohammed, Sergio Cadoni, Antipova Mariya, Sauid Ishaq, Toshio Kuwai

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Abstract Body: BACKGROUND AND AIMS: Scissor-type knives have shown safety and efficacy in endoscopic submucosal dissection (ESD) procedures, particularly in studies from Japan. However, the safety and efficacy of these devices in international settings, particularly outside Japan is not well established. Methods: This is prospective, multicenter, observational study conducted across nine international centers, including Japan, China, Italy, UK, and Russia, encompassing a total of 461 lesions from 460 patients. Four Japanese institutions contributed 162 lesions (Japanese institutions group, JAG), and five institutions outside Japan contributed 299 lesions (non-Japanese institutions group, NJAG). Propensity score matching was performed at a 1:1 ratio, resulting in 120 matched lesion pairs. A Key outcomes were then compared between groups.The primary outcome was the incidence of intraoperative perforation, defined as the visualization of intraperitoneal adipose tissue due to injury to the muscle layer during the procedure. Secondary outcomes included procedure time, resection outcomes (en bloc, histological complete, and curative resection rates), and the incidence of post-procedural adverse events such as delayed bleeding and perforation. Results: The overall perforation rate during ESD procedures was 0.87%. Intraoperative perforations were observed more frequently in NJAG than JAG (3 vs 1 event, 1.9% vs. 0.33%, respectively), though this difference was not statistically significant ($p = 0.127$). The overall incidence of delayed bleeding was also 0.87%, with no delayed bleeding reported in NJAG. Post propensity matching analysis revealed a significantly slower median resection speed in NJAG compared to (9.12, 0.86–56.57 vs 26.21, 1.95–93.54 mm2/min, $p<0.001$). Both histological complete resection and curative resection rates were significantly lower in NJAG than in JAG with rates of 88.3% vs 98.3% for histological complete resection and 83.3% vs 95% for curative resection (both $p < 0.01$). In the multivariate analysis including all lesions, no significant risk factors were identified for intraoperative perforation, delayed perforation, or delayed bleeding, while several identified risk factors with incomplete resection were identified. The OR for these factors were as follows: Institution outside Japan: OR = 8.435 (95% CI: 2.487–28.613, $p < 0.001$); Slower resection speed: OR = 5.204 (95% CI: 1.095–24.734, $p < 0.05$); Severe submucosal fibrosis: OR = 7.657 (95% CI: 2.355–24.891, $p < 0.01$); Pathology of T1b: OR = 4.508 (95% CI: 1.517–13.398, $p < 0.01$). Conclusions: The use of scissor-type knives in colorectal ESD outside Japan demonstrated a favorable safety profile. However, certain performance outcomes, such as resection speed and resection success rates, were inferior compared to Japanese institutions.

ID: 4256324

EVALUATING RECURRENCE AND PROGNOSIS IN LOW-RISK T1B COLORECTAL CANCER: A MULTICENTER RETROSPECTIVE STUDY

Hirohito Tanaka, Shiko Kuribayashi, Takanari Masuo, Masanori Sekiguchi, Atsuo Iwamoto, Yoko Hachisu, Tetsuo Nakayama, Yuko Kimura, Tomoyuki Masuda, Kazuhiro Takahashi, Kyoko Marubashi, Yoji Takauchi, Toshio Uraoka

ASGE Oral/Poster: Disclosure Compliance: I understand. DDW 2025 PARTICIPANT DISCLOSURE FORM: Hirohito Tanaka: NO financial relationship with a commercial interest | Shiko Kuribayashi: NO financial relationship with a commercial interest | Takanari Masuo: No Answer. | Masanori Sekiguchi: No Answer. | Atsuo Iwamoto: No Answer. | Yoko Hachisu: No Answer. | Tetsuo Nakayama: No Answer. | Yuko Kimura: No Answer. | Tomoyuki Masuda: No Answer. | Kazuhiro Takahashi: No Answer. | Kyoko Marubashi: No Answer. | Yoji Takauchi: No Answer. | Toshio Uraoka: No Answer.

Abstract Body: Background: Risk factors for lymph node metastasis (LNM) after endoscopic resection (ER) for T1 colorectal cancer (CRC) include submucosal (SM) invasion ≥1000μm, lymphatic invasion (Ly1), venous invasion (V1), poorly differentiated adenocarcinoma, signet-ring cell carcinoma, mucinous carcinoma, tumor budding (BD2/3), and positive vertical margin (VM1). Surgical resection with lymph node dissection is recommended when any of these factors are present. However, cases with SM invasion ≥1000μm alone as a risk factor are recently suggested to have a low risk of LNM. While several studies have reported on this topic, they were limited to specialized centers. The aim of this study was to investigate the LNM rate, recurrence rate and prognosis of T1 CRC cases classified as “Low-risk T1b”(SM invasion ≥1000μm without other risk factors), treated with either endoscopic resection (ER) or surgical resection (SR) based on clinical data from regional general hospitals. Methods: We retrospectively analyzed 686 cases of T1 CRC treated with ER or surgical resection (SR) between 2009 and 2019 at our institution and nine affiliated regional hospitals. Among these, 217 cases with complete data on risk factors for LNM were included in the analysis (105 ER and 112 SR cases). LNM rate, recurrence rate and prognosis, focusing on "Low-risk T1b", were evaluated. Results: